§2.2: andwiched Singularities, Extended Graphs, and Decorated Germs

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August 18, 2026

1. Proposition IV (as seen from the paper)

Proposition 1.1. *Let $(X, 0)$ be a rational singularity with reduced fundamental cycle, and $\mathcal{C}$ a plane curve germ corresponding to $(X, 0)$. If $\mathcal{C}$ has smooth branches, the number of branches is given by* $\operatorname{mult} X - 1$.

2. Definitions

Definition 2.1. Let $(X, 0) \subset (\mathbb{C}^N, 0)$ be a surface singularity. The *multiplicity* $\operatorname{mult} X$ is defined geometrically as the number of intersections $\#X \cap L$ of X with a generic complex $(N-2)$-dimensional affine subspace $L \subset \mathbb{C}^N$ passing close to the origin. For rational singularities, multiplicity is a topological invariant computed from the resolution graph by the formula:

$$\operatorname{mult} X = -Z_{min}^2$$

The components of this definition are broken down as follows:

- **$(X, 0) \subset (\mathbb{C}^N, 0)$:** A complex surface inside an N-dimensional complex space with an isolated singularity at the origin.
- **$\operatorname{mult} X$:** A geometric measure of the folding or density of the surface at the singular point (a smooth point has a multiplicity of 1).
- **L:** A generic complex linear subspace of codimension 2. Its intersection with the 2-dimensional surface X yields a finite, countable set of isolated points.
- **"Passing close to the origin":** Perturbing L slightly away from the origin unfolds the singularity. This allows the distinct generic intersection points ($\#X \cap L$) to be counted cleanly.
- **Z_{min}:** The fundamental (minimal) cycle. This is the unique minimal positive integer combination of exceptional curves $Z = \sum a_i E_i$ satisfying $Z \cdot E_i \leq 0$ for all components E_i in the resolution graph.
- **$-Z_{min}^2$:** The negative self-intersection of the fundamental cycle. Because the resolution intersection matrix is negative definite, Z_{min}^2 is negative. Flipping the sign yields a positive integer matching the geometric intersection count.

For a branch of a plane curve germ $\mathcal{C}$ to be *smooth*, it must satisfy the following conditions:

- **No discontinuities:** The branch is topologically connected and forms a continuous path through the origin.
- **No sharp edges or kinks:** Geometrically, the branch possesses a well-defined, continuously turning tangent vector at every point, including at the origin.
- **Analytic smoothness:** Algebraically, each individual branch can be parameterized locally by a single smooth complex parameter $t \mapsto (x(t), y(t))$ such that the derivative vector is non-vanishing, meaning the branch is locally isomorphic to a non-singular complex disk.

Artin's Theorem and the Genus Formula

To prove our result about the minimal resolution graph G, we need to check the arithmetic genus.

Theorem 3.1 (Artin's Characterization Theorem)**.** *Let $(X, 0)$ be a normal surface singularity, and let Z be its fundamental cycle on a good resolution $\tilde{X}$. The singularity is rational if and only if the arithmetic genus of the fundamental cycle is zero:*

$$p_a(Z) = 0$$

Proposition 3.2 (The Arithmetic Genus Formula)**.** *For any effective cycle $Y = \sum_{i=1}^n c_i E_i$ on a resolution surface, its arithmetic genus $p_a(Y)$ is calculated from the graph data by:*

$$p_a(Y) = 1 + \frac{1}{2}Y^2 + \frac{1}{2}\sum_{i=1}^{n} c_i\big(2g(E_i) - 2 - E_i^2\big)$$

Proof of Proposition 2.4

Proof. Let $(X, 0)$ be a rational surface singularity. We are given that its fundamental cycle $Z_{\min}$ is reduced. Recall that being reduced means all the coefficients c_i in the cycle are exactly 1:

$$Z_{\min} = \sum_{v \in G} E_v$$

where G is the minimal resolution graph with n vertices. Recall that since our exceptional curves E_v are rational curves, their geometric genus is zero ($g(E_v) = 0$) for every vertex $v \in G$.

Now we plug $c_v = 1$ and $g(E_v) = 0$ into the arithmetic genus formula for $Z_{\min}$:

$$p_a(Z_{\min}) = 1 + \frac{1}{2}Z_{\min}^2 + \frac{1}{2}\sum_{v \in G}\big(2(0) - 2 - E_v^2\big)$$

$$p_a(Z_{\min}) = 1 + \frac{1}{2}Z_{\min}^2 + \frac{1}{2}\sum_{v \in G}(-2 - v \cdot v)$$

$$p_a(Z_{\min}) = 1 + \frac{1}{2}Z_{\min}^2 - n - \frac{1}{2}\sum_{v \in G}(v \cdot v)$$

Because $(X, 0)$ is a rational singularity, Artin's theorem states that $p_a(Z_{\min}) = 0$. Let's set the equation to zero and multiply everything by 2 to clear the fractions:

$$0 = 2 + Z_{\min}^2 - 2n - \sum_{v \in G}(v \cdot v) \implies Z_{\min}^2 = \sum_{v \in G}(v \cdot v) + 2n - 2$$

Recall that for any tree graph with n vertices, the sum of all vertex valences $d(v)$ is exactly equal to twice the number of edges. Since a tree has $n - 1$ edges, the sum of the valences is $2(n - 1) = 2n - 2$. We can substitute $\sum_{v \in G} d(v) = 2n - 2$ into our formula:

$$Z_{\min}^2 = \sum_{v \in G}(v \cdot v) + \sum_{v \in G} d(v) = \sum_{v \in G}\big(v \cdot v + d(v)\big)$$

Next, we look at the conditions from **Proposition 2.2**. The plumbing graph G' for the smooth point is made by adding m branches as (-1) leaf vertices to G. Proposition 2.2 tells us that:

1. There is exactly one special vertex $v_0' \in G'$ where $v_0' \cdot v_0' + a(v_0') = -1$.
2. For all other vertices $v' \neq v_0' \in G'$, $v' \cdot v' + a(v') = 0$.

When we go from G' back to our minimal graph G by removing the m leaf branches, the vertex valences change:

- For any vertex $v \neq v_0'$, no branches were attached there, so the valence in G is the same as in G' ($d(v) = a(v)$). This means $v \cdot v + d(v) = 0$.
- For the special vertex v_0', all m branches were attached here, so its valence drops by m in the minimal graph ($d(v_0') = a(v_0') - m$).

Now we can compute the sum for Z_{min}^2 using these two facts:

$$Z_{\text{min}}^2 = \sum_{v \neq v_0'} (0) + \big(v_0' \cdot v_0' + d(v_0')\big)$$

$$Z_{\text{min}}^2 = v_0' \cdot v_0' + a(v_0') - m$$

Using the condition $v_0' \cdot v_0' + a(v_0') = -1$ from Proposition 2.2, this simplifies directly to:

$$Z_{\text{min}}^2 = -1 - m$$

Recall that the multiplicity of a rational singularity is defined as $\operatorname{mult} X = -Z_{\text{min}}^2$. Therefore:

$$\operatorname{mult} X = -(-1 - m) = m + 1$$

If we rearrange this equation to solve for the number of branches m, we get:

$$m = \operatorname{mult} X - 1$$

This completes the proof. □

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